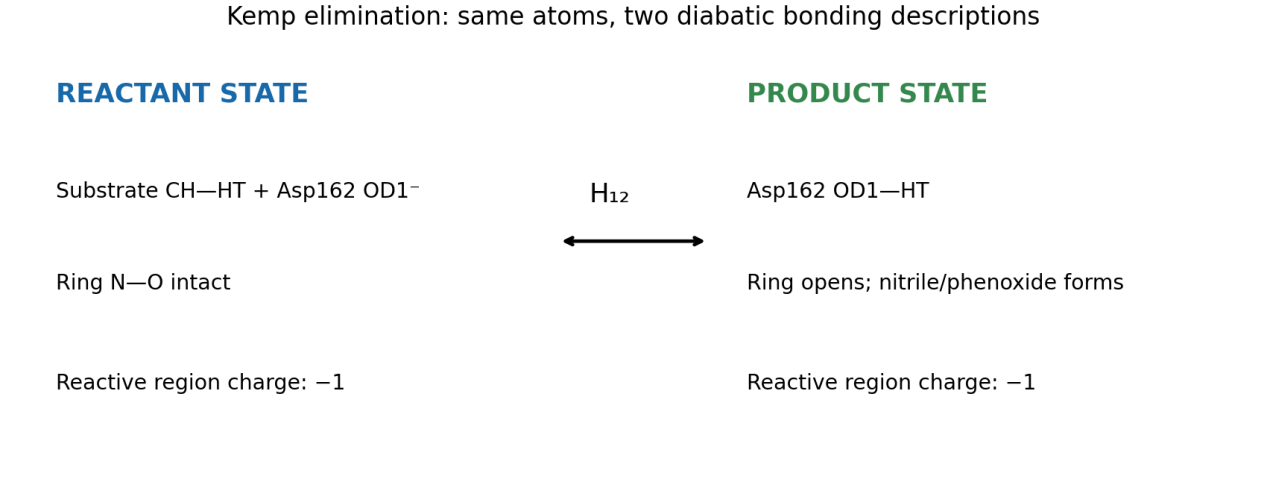
EVB with Kemp elimination: one method, two engines

- 30-minute talk, then30-minute practical.
- Understand states, coupling, offsets and mapping/FEP.
- Compare the paper's Q6 calculation with an OpenMM adaptation.
- Today: saved results and target-matching exercises, not a validated reproduction.

Define the Kemp reaction and mapped states

- Asp162 abstracts the proton from5-nitrobenzisoxazole.
- CH-HT and OR-NR break; Asp OD1-HT forms.
- Other bonded terms, charges and nonbonded parameters also change.
- R/P share the same atoms; the22-atom reactive region has charge −1 in both.



What is EVB? What is a diabatic state?

- EVB describes a reaction by mixing classical bonding/charge patterns.
- A diabat is one force-field state, not a second set of atoms.
- Each state defines bonded terms, charges, LJ and exclusion rules.
- Reference calibration and ensemble sampling are needed for free energies.

The two-state EVB Hamiltonian

- H = [[V_R, H12], [H12, V_P + alpha]].
- Its lower eigenvalue is the coupled ground-state potential.
- Eigenvector weights describe state mixing.
- The same matrix algebra applies in Q6 and OpenMM.

Coupling, offset and barrier are different quantities

- Alpha sets the relative diabatic energy origin.
- H12 mixes the states and opens an avoided crossing.
- Neither coefficient is the activation barrier.
- Fit independent reference evidence for a physical model; target matching today is a labelled exercise.

What are Q6, OpenMM and EVBOpenMM?

- Q6: standalone MD/EVB toolkit, including Qdyn dynamics and QFEP analysis.
- OpenMM: programmable MD engine with Python APIs and CPU/GPU kernels.
- EVBOpenMM: the added EVB force/model/workflow layer used here.
- File formats, analysis, boundaries and precision can differ even when EVB theory is shared.

Mapping/FEP and profile reconstruction

- Sample lambda*V_R + (1-lambda)*V_P at several mapping windows.
- Save both state energies at every sampled configuration.
- Mix energies per frame, then reweight and bin consistently.
- Lambda is not physical reaction time or the adiabatic state-mixing weight.

What the Nature paper simulated

- Conventional MD: Amber ff19SB/GAFF2 with OPC.
- EVB: Q6, OPLS-AA, TIP3P, SCAAS sphere and local reaction field.
- 30 replicas/case;30 ns equilibration;51 windows x200 ps production.
- Des27.7-IN computed mean13.2 +/-0.6 kcal/mol; uncertainty is replica SD.

What this OpenMM example implements

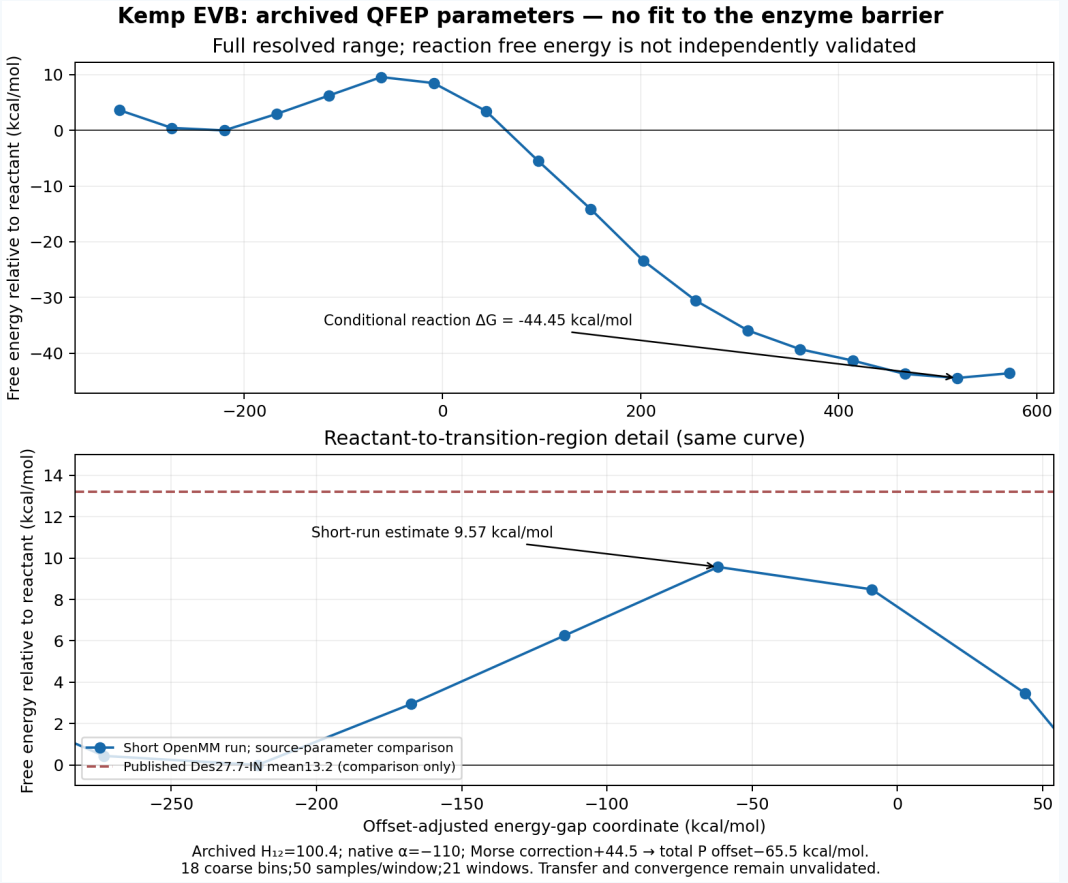
- All6330 atoms and the supplied reactive changes are imported for Des27.7-IN.
- Workflow: minimization, heating, equilibration and restartable mapping production.
- Boundary changes: direct nonperiodic electrostatics, droplet wall and fixed outer protein.
- No Q6 LRF equivalence or converged physical barrier is claimed.

Recovered parameters and energy conventions

- Archived qfep.inp: constant H12=100.4 and native alpha = -110 kcal/mol.
- Reactive FEP sections match the Des27 inputs after Q-index mapping.
- Q6 Morse minimum=-De; EVBOpenMM minimum=0.
- +44.5 on P preserves the relative convention: total P offset=-65.5 kcal/mol.

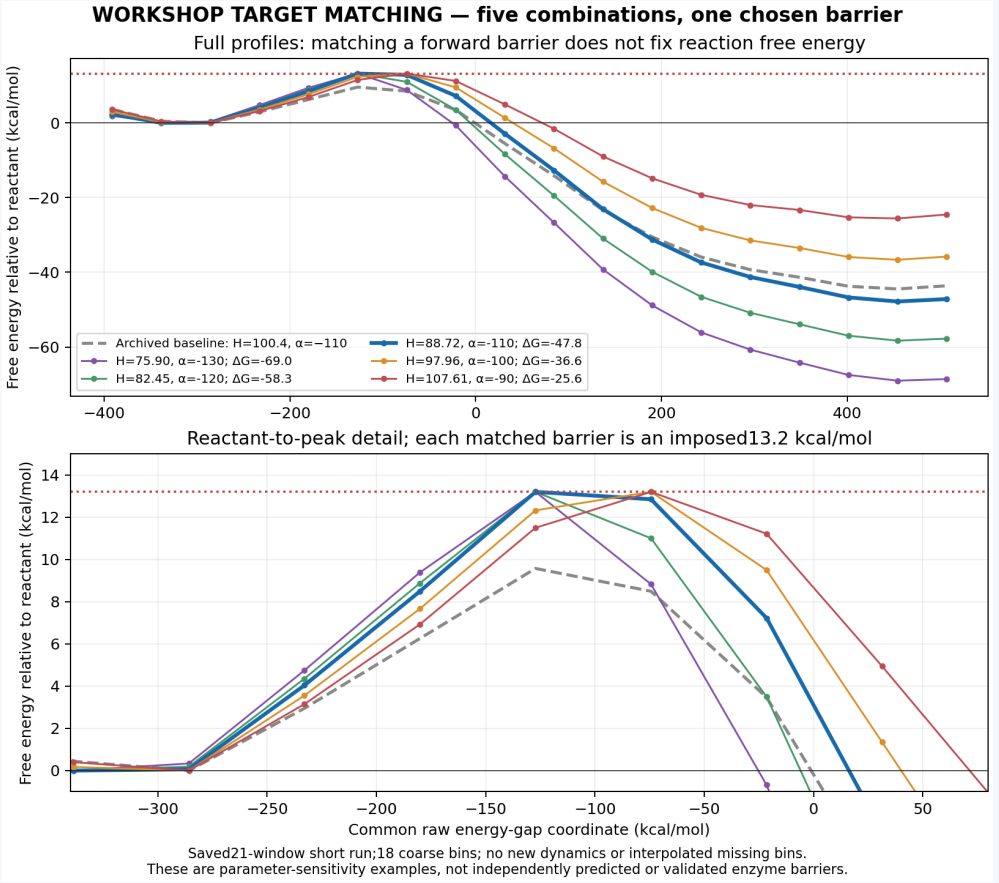
Read the source-parameter comparison

- Saved21-window short OpenMM run;50 records/window.
- At18 coarse bins the source-based comparison is about 9.57 kcal/mol.
- No fit to13.2 was used for this curve.
- Changed boundary, short correlated data and coarse binning limit interpretation.



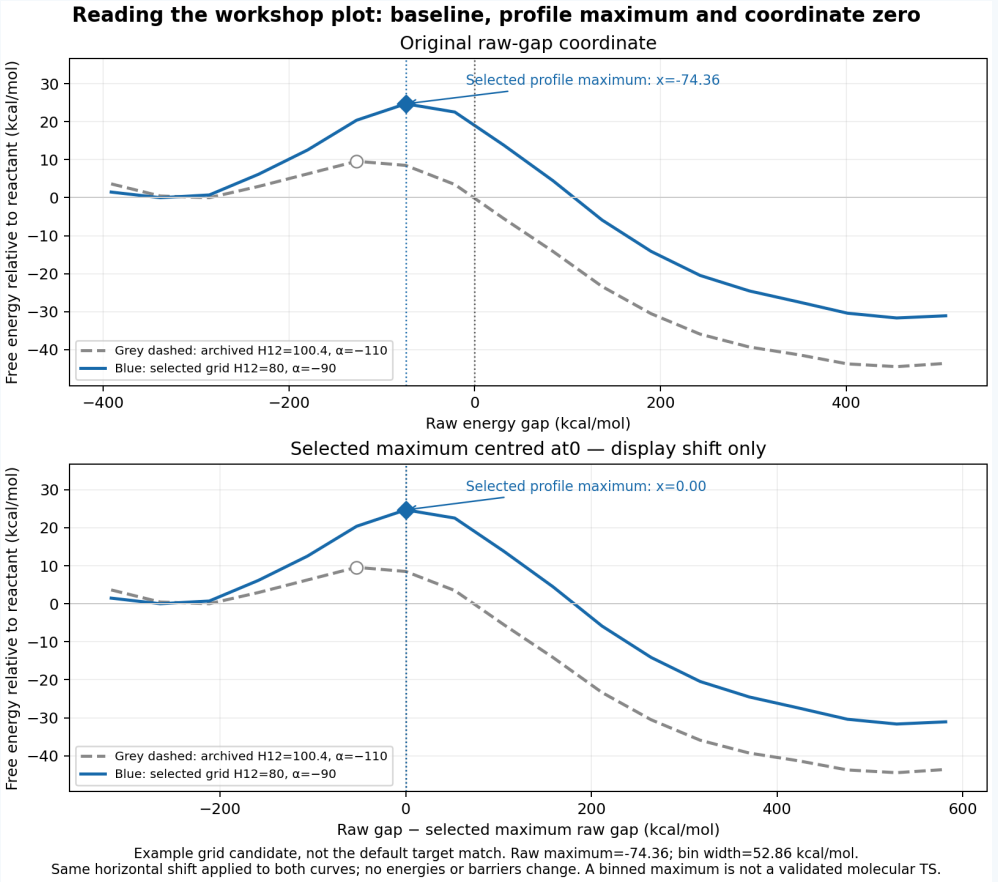
Many parameter pairs can match one target

- Reuse the saved energies: no additional MD for the workshop.
- 25 grid combinations plus5 examples matching the chosen13.2 barrier.
- One example: H12 about 88.72, native alpha = -110 kcal/mol.
- Matching pairs have different reaction free energies; this is not unique physical calibration.



Grey baseline, profile maximum and coordinate zero

- Grey dashed: archived baseline on the same energies; blue: selected candidate.
- The diamond marks a binned maximum, not a validated molecular TS.
- Raw or corrected gap zero need not be a free-energy maximum.
- TS-centred view applies the same display shift to both curves; energies/barriers do not change.



Practical: first 15 minutes

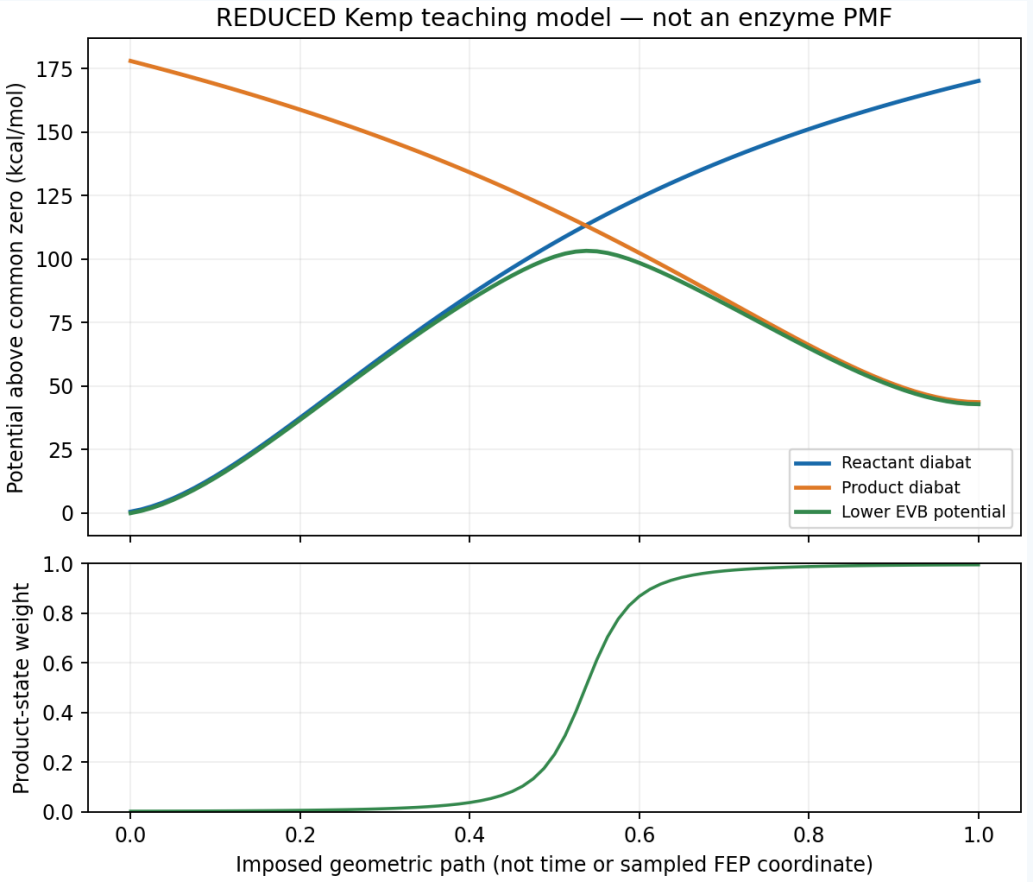
- 0-5: inspect the reaction and Q atom mapping.
- 5-10: run the core or open its saved viewer; vary coupling/offset.
- 10-15: read the archived input and recompute the source profile.
- Use run_practical.py or the offline index; the sample energies are included.

Practical: next 15 minutes

- 15-25: compare parameter pairs matching one barrier; record reaction free energies.
- 25-30: select H12=80, alpha = -90 and toggle the common display shift.
- Identify grey baseline, blue maximum marker and coordinate zero.
- Discuss one shared EVB concept and one Q6/OpenMM implementation difference.

Backup: five-atom potential exercise

- Three native Morse terms plus selected C-N harmonic terms.
- 81 imposed geometries; illustrative H12/extra offset.
- OpenMM lower eigenvalues checked against EVBOpenMM.
- Enzyme, solvent and most force-field terms are omitted; this is not a PMF.

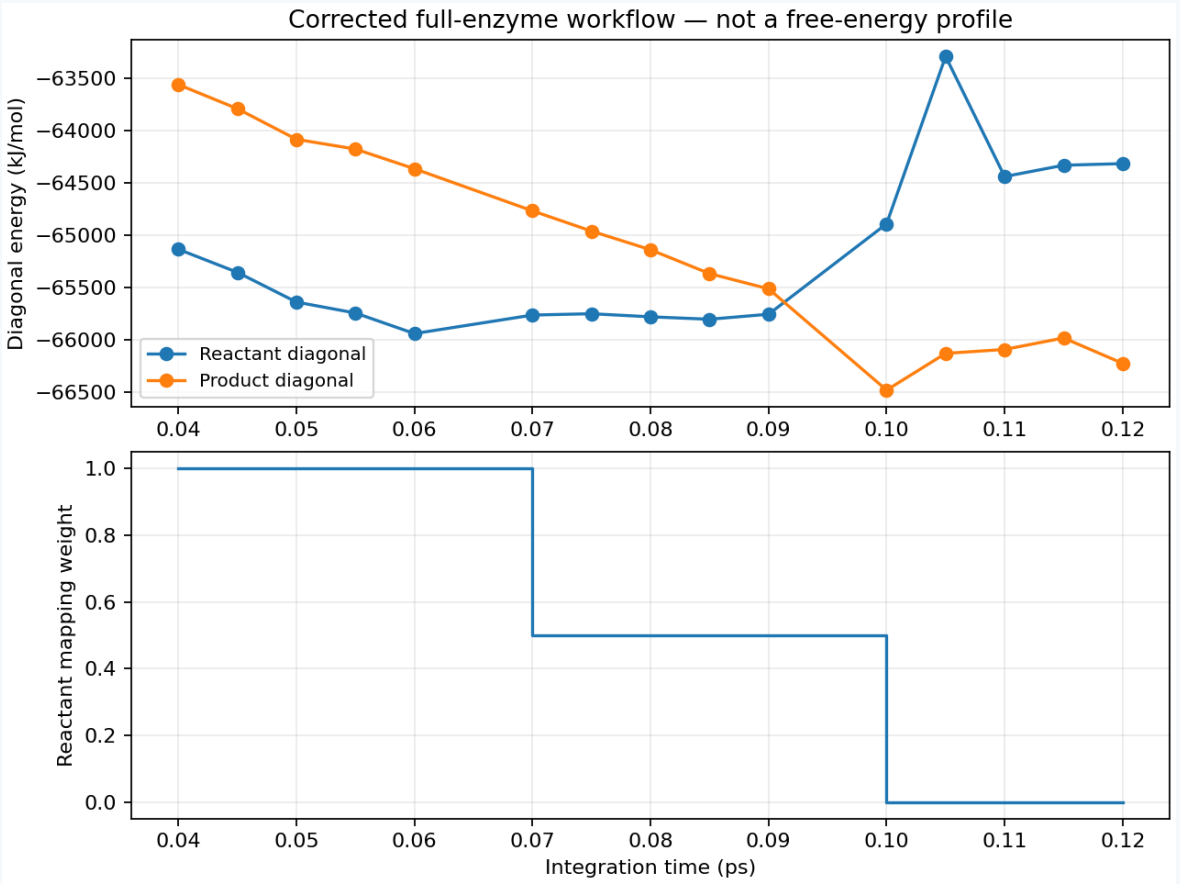


Backup: native Morse energy origins

- Q6: D*(exp(-2*a*dr)-2*exp(-a*dr)), with minimum-De.
- Zero-minimum Morse adds De for every active bond.
- For these terms R gains182.75, P gains138.25 kcal/mol.
- P-R shifts by-44.5, so add+44.5 to P for convention equivalence.

Backup: full-enzyme execution evidence

- 6330 atoms; corrected native LJ import.
- Minimization, heating, equilibration and three production windows executed.
- 480 MD steps at 0.25 fs; checkpoint pause/resume tested.
- This saved DCD is a0.12 ps engineering test, not an observed reaction.



Backup: source-profile analysis order

- Native QFEP input supplies100.4 coupling and-110 state offset.
- Constant coupling permits reanalysis from the saved state energies.
- Evaluate the lower eigenvalue per frame before exponential bin averaging.
- The sample cannot support native skip4000/minimum 1000 settings; actual settings are disclosed.

Backup: how to run longer simulations

- Change equilibration_steps and window_equilibration_steps in the chosen JSON.
- production_steps_per_window and lambda_schedule set the production length.
- Use a new output directory for changed settings and a separate replica seed.
- The long template collects10.2 ns; CPU runtime can greatly exceed one night.

Backup: Amber does work with OpenMM

- OpenMM reads Amber prmtop/inpcrd files.
- AmberTools is needed to rebuild the optional Amber lane, not this practical.
- That classical compatibility example is separate from OPLS EVB.
- No Q6, Rosetta or Voth/LAMMPS build is required for today.

Paper, supporting information and deposited inputs

- Listov et al., Complete computational design of high-efficiency Kemp elimination enzymes, Nature (2025).
- Article: doi:10.1038/s41586-025-09136-2; associated Supplementary Information, Table 3.
- Simulation inputs: Zenodo doi:10.5281/zenodo.14563437 (EVB_Simulations.zip).
- Design code: Fleishman-Lab/denovoKemp; design code and EVB inputs have different roles.

Related protocol evidence: do not conflate the datasets

- Gutierrez-Rus et al., JACS (2025), doi:10.1021/jacs.4c09428, with its Supporting Information.
- GNCA Kemp inputs: Zenodo doi:10.5281/zenodo.12666978; Kemp_variants.zip.
- Recovered qfep.inp supplies H12=100.4 and alpha=-110 kcal/mol.
- GNCA is a different scaffold: these are protocol evidence, not replacement Des27 coordinates or proof of parameter transfer.

Atom names: read the actual mapping

- Actual PDB identities: SUB257 and catalytic ASP162; stale FEP comments use different residue numbers.
- CH=proton-bearing substrate carbon; HT=transferred H; NR=ring nitrogen; OR=ring oxygen.
- Asp OD1 accepts HT; OD2 is the other oxygen; CG is its carboxyl carbon.
- Q indices are local reactive indices; PDB serials are global atom identifiers. Atom names are not force-field types.

Detailed Kemp mechanism and state bookkeeping

- R: CH-HT and OR-NR bonded; substrate neutral, Asp carboxylate -1.
- Coupled proton abstraction and ring opening: OD1 accepts HT, CH-NR gains nitrile character, OR gains phenoxide character.
- P: OD1-HT bonded; CH-HT and OR-NR absent; nitrile/phenoxide product -1 and neutral AspH.
- Total reactive charge stays -1. This two-state model introduces no separately sampled carbanion intermediate.

Published Q6 versus our short OpenMM example

- SI Table3: Des27.7-IN Q6 barrier=13.2 +/-0.6 kcal/mol (replica SD).
- Short archived-parameter OpenMM estimate=9.57; no independent-replica uncertainty.
- Difference=-3.63 kcal/mol; different boundary and limited sampling prevent an agreement claim.
- A target-matched 13.2 workshop curve is not independent reproduction.

